NVIDIA | AIStore

MLPERF STORAGE V3.0 · CLOSED SUBMISSION

AIStore on Amazon Web Services

Three-node AWS shared object storage with five benchmark clients

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aistore-3n-aws-i8ge-150g-5client-m8gn24xl
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SUBMITTER
NVIDIA

SUBMISSION DATE
July 24, 2026

BENCHMARK
MLPerf Storage v3.0
mlpstorage 3.0.40

DIVISION
CLOSED

SOLUTION TYPE
Open-source shared object storage

DEPLOYMENT
Cloud · Kubernetes

POWER REQUIREMENTS
Provider managed; cloud exemption applies

SUBMITTED WORKLOAD
RetinaNet training

5 benchmark clients
m8gn.24xlarge

→

3-node AIStore cluster
3 proxies · 3 targets · 24 target NVMe devices

System topology and measured data path

Clients access AIStore exclusively through its S3-compatible API. A proxy admits each request and redirects the object transfer to the responsible target.

5 × m8gn.24xlarge benchmark clients
Open MPI · DLIO · s3dlio · equal rank placement per participating host

↓ S3 over HTTP through round-robin across supplied proxy endpoints

3 × AIStore proxy pods
Request admission, cluster map, placement, and HTTP redirect

↓ direct client-to-target redirected object transfer

3 × AIStore target pods on i8ge.24xlarge
24 local NVMe devices · one native distributed object namespace

Architecture reference
For AIStore proxy, target, routing, and deployment details, see the AIStore architecture overview.

Cloud infrastructure inventory

5
BENCHMARK CLIENTS

3
AISTORE STORAGE NODES

24
TARGET NVME DEVICES

Component	Benchmark clients	AIStore storage nodes
Quantity / role	5 external MPI hosts	3 Kubernetes nodes; one proxy and one target per node
Cloud shape	m8gn.24xlarge	i8ge.24xlarge
Compute / network	1 × AWS Graviton4; 96 captured cores; 384 GB captured memory; 300 Gb/s data networking	1 × AWS Graviton4; 96 captured cores; 768 GB captured memory; 150 Gb/s data networking
Local media	Present but not used for MLPerf data	8 × 7.5 TB TLC NVME per target; 24 devices total
Platform	Amazon Linux 2023	Kubernetes 1.34.2

Client-role clarification
Client root volumes were encrypted gp3. Client-local storage was not used for benchmark data. All submitted objects were stored by the AIStore targets.

Storage capacity

- 180.00 TB aggregate target-drive nameplate capacity.
- Native AIStore bucket on local target NVMe.
- Mirroring and erasure coding disabled.

Cloud-managed facilities

- Physical switching, racks, cooling, and PSUs are provider managed.
- The MLPerf cloud-deployment power and rack-unit exemption applies.

Reproducible software stack

Layer	Version / configuration used	Reference
AIStore	v4.8, commit `42242b334fd57e993919f04acf3c2011cf2db675`	source
MLPerf Storage	mlpstorage 3.0.40; captured code image(s) `7eb38695e3d8e944b10e6601f76965dc`	19e9f1666e9fe194fb69a6f8f34f53fbe34fe4a2
I/O stack	DLIO with s3dlio; path-style S3 over HTTP	DLIO
Client platform	Amazon Linux 2023	AWS m8gn.24xlarge clients / i8ge.24xlarge storage
Storage platform	Kubernetes 1.34.2	AWS m8gn.24xlarge clients / i8ge.24xlarge storage

Object path

- Native AIStore bucket through the S3-compatible API.
- Path-style S3 and standard AIStore HTTP redirects.
- Recorded endpoint selection: round-robin across supplied proxy endpoints.

MPI placement

- Equal ranks per participating client.
- `--bind-to none --map-by node`.
- Client-local NVMe outside the data path.

Captured non-default S3 and TCP settings

`S3DLIO_FOLLOW_REDIRECTS=1` `S3DLIO_RANGE_THRESHOLD_MB=1024` `S3DLIO_MULTIPART_THRESHOLD_MB=4096` `S3DLIO_OPERATION_TIMEOUT_SECS=300` `S3DLIO_CONNECT_TIMEOUT_SECS=60` `S3DLIO_MAX_RETRY_ATTEMPTS=10` `S3DLIO_POOL_MAX_IDLE_PER_HOST=64` `AWS_MAX_ATTEMPTS=10` `AWS_RETRY_MODE=standard` `net.ipv4.tcp_tw_reuse=1` `net.ipv4.tcp_retries2=15`

The matching YAML is the authoritative per-host hardware, operating-system, network, environment, sysctl, drive, and capability inventory. Secret values are redacted.

CLOSED workload configuration

This page records the submitted workload parameters and a compact snapshot of the validated result artifacts.

Parameter	Submitted configuration
Model / accelerator	RetinaNet / B200
Scale	255 ranks; 51 per client
Dataset	31,917,318 JPEG objects across 3,192 key-prefix subfolders; 9,600 GiB
Reader	batch size 24; reader.read_threads=1; S3 prefetch window=16
Invocation	8 epochs; one warm-up plus five measured runs

Validated result snapshot

Workload	Mean AU	Samples/s	Training I/O
RetinaNet	88.77%	114,053.25	34.30 GiB/s

Interpretation
The mean AU, samples/s, and I/O values are the aggregate values reported by the validated MLPerf artifacts. The corresponding packaged result files remain authoritative.

Primary sources and submission scope

These links connect the tested configuration to product specifications, source code, and protocol behavior.

NVIDIA AIStore documentation
https://docs.nvidia.com/aistore

AIStore architecture and deployment portability
https://docs.nvidia.com/aistore/overview

AIStore S3 compatibility
https://docs.nvidia.com/aistore/s3compat

AIStore open-source repository
https://github.com/NVIDIA/aistore

Cloud instance specification
https://aws.amazon.com/ec2/instance-types/

MLCommons Storage benchmark and rules
https://github.com/mlcommons/storage

Exact AIStore source revision
42242b334fd57e993919f04acf3c2011cf2db675

Document scope
This PDF describes only `aistore-3n-aws-i8ge-150g-5client-m8gn24xl` and the result artifacts stored under that system ID. Other client counts, workload configurations, software revisions, and clouds have separate same-name YAML/PDF descriptions.

Machine-readable companion

The same-name YAML is the authoritative structured system inventory.

Validated artifacts

Logs, metadata, resolved configurations, summaries, code-image markers, and aggregate results remain in the matching results directory.

Important context and disclosures

- This cluster can be pushed closer to its aggregate nominal network line rate. For this submission, we chose settings that kept accelerator utilization (AU) compliant and performance stable across all six runs.
- The AWS UNet3D and RetinaNet submissions use different client counts, rank counts, reader settings, object profiles, and achieved I/O rates. They are valid workload-specific points, not evidence that either configuration fully saturates the AWS platform.
- OCI and AWS three-node results are not like-for-like cloud comparisons. Captured storage/client networking is 100/100 Gb/s on OCI and 150/300 Gb/s on AWS; shapes, client counts, datasets, and workload tuning also differ.
- This submission used HTTP to keep the benchmark path simple and consistent. AIStore supports HTTPS/TLS. With HTTPS, AIStore disables the file-to-socket sendfile fast path (encrypted responses use buffered transfer, and TLS encryption and handshakes add CPU work), so HTTPS performance may differ.
- NVIDIA AIStore is not a managed service offering. It is open source, and its source code, deployment guidance, and deployment recipes are publicly available. These are deployment-specific measurements, not performance guarantees or competitive claims.

Supporting documentation

The following documents provide disclosure, scaling, and deployment context applicable to this NVIDIA AIStore system.

AIStore Disclosure

AIStore Scaling and Cloud Portability